Fiscal Year:	FY 2018	**Task Last Updated:**	FY 11/06/2017
PI Name:	Turner, Nancy Ph.D.		
Project Title:	Radiation-Induced Apoptosis Avoidance and Colon Tumorigenesis: Epigenetic Regulation in Adult Stem Cells		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	PI moved in mid-2018 to Michigan State University from Texas A&M University AgriLife Research (Ed., Jan 2019)		
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-RADIATION
Start Date:	01/07/2015	**End Date:**	06/20/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/20/2018 due to PI move to Michigan State University from Texas A&M (Ed., 7/15/19) NOTE: End date changed to 1/6/2019 per S. Monk/HRP-LaRC (Ed., 12/21/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Carroll, Raymond Ph.D. (Texas A&M University) Chapkin, Robert Ph.D. (Texas Agricultural Experiment Station) Ford, John Ph.D. (Texas Engineering Experiment Station)		
Grant/Contract No.:	NNX15AD64G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	One key hallmark in carcinogenesis that can occur in cells surviving irradiation is the evasion of apoptosis, which may develop in part from altered gene expression. We have shown that the microbial metabolite butyrate and long-chain omega-3 fatty acids impact expression of apoptosis related genes through epigenetic mechanisms (e.g., DNA methylation and histone acetylation status). We also detected changes in colon microbiota and their metabolites after radiation, suggesting these changes may be upstream of the epigenetic changes found in irradiated colonocytes. The recent discovery of UHRF1, which interacts with and regulates the activity of DNA methyltransferases and histone deacetylases, may be a pivotal regulatory element underlying control of apoptosis induction. Importantly, UHRF1 also reads DNA methylation marks and facilitates the inheritance of DNA methylation states to daughter cells during cell proliferation. If colon adult stem cells are especially sensitive to radiation induced changes in gene expression, epigenetic state, and DNA-damage induced apoptosis, then a relatively rapid expansion of the pool of preneoplastic cells would occur and thus contribute to an increased risk of carcinogenesis. Colon adult stem cell marker mice make it possible to specifically monitor physiological responses to radiation exposure in these critical cells. Therefore, the goal of this work will be to determine the impact of Si, Ti, or Fe ions on the regulation of colonocyte apoptosis in mice exposed to doses representative of the total doses on long-duration missions to Mars or asteroids (0.10, 0.25, and 0.50 Gy). It will address the epigenetic mechanisms whereby apoptosis is avoided by colonoctyes overall, but importantly it will define how adult stem cells respond to radiation. Finally, we will employ a dietary countermeasure to alter colon microbiota and butyrate concentrations in order to determine if the down stream changes in epigenetic effectors caused by the diet will reduce or eliminate the negative effects of radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The colon is proposed to be the third most susceptible organ-site to radiation-induced tumorigenesis, which means that patients undergoing abdominal radiotherapy protocols are at risk of developing secondary tumors there. The enhancement of risk may result from perturbations in the microenvironment that creates a putative cancer niche that includes the microbiota, their metabolites (e.g., the short chain fatty acids), and the adult colon stem cells. We have demonstrated that a diet containing a fermentable fiber along with omega-3 polyunsaturated fatty acids protects against colon cancer in part through the induction of apoptosis. Our prior work also documents the ability of microbial metabolites to alter expression of apoptosis related genes through their impact on epigenetic state. For the current work, we have adopted a systems biology approach and are studying colon epithelial cells, the colon microbiota and their metabolites to more fully appreciate the impact radiation on colon health. Data derived from this study will contribute to our understanding of the processes that contribute to radiation-induced tumorigenesis and provide potential dietary countermeasures to those effects.
Task Progress:	Introduction: The complex radiation signature of the space environment poses many challenges, including contributing to carcinogenesis. Apoptosis evasion is a characteristic of cells post irradiation. The colon is proposed to be the third most susceptible organ-site to radiation-induced tumorigenesis, possibly because of perturbations in the microenvironment that creates a putative cancer niche. The colonic microenvironment includes the microbiota and their metabolites, such as the short chain fatty acids. We have demonstrated that a diet containing a fermentable fiber along with omega-3 polyunsaturated fatty acids protects against colon cancer in part through the induction of apoptosis. Our prior work also documents the ability of microbial metabolites to alter expression of apoptosis related genes through their impact on epigenetic state. Because transformation of adult stem cells is an extremely efficient route towards initiating intestinal cancer, we are using the Lgr5-EGFP-IRES-creERT2 knockin mice to study the impact of radiation exposure on intestinal stem cells and to compare their responses to those of differentiated cells. This model allows us to visualize and isolate adult intestinal stem cells, in order to compare their physiological and epigenetic responses to those of differentiated colonocytes after low dose radiation exposure. We have adopted a systems biology approach and are studying colon epithelial cells, the colon microbiota and their metabolites to more fully appreciate the impact radiation on colon health. Materials and Methods: To conduct these experiments we are using a transgenic mouse line that makes visualization of colon stem cells possible. The mice are bred, genotyped, and raised until they reach an average age of 40 weeks before starting the treatments, in order to simulate responses that would occur in a 35-55 year old astronaut. In each experiment, a set of sham-irradiated mice is included, which serve as controls. Three weeks before irradiation all mice are weighed and randomly allocated to treatment groups so that initial weights, ages, and gender distributions are similar among the groups. The radiation exposures at Brookhaven National Laboratory (BNL) include 0, 10, 25, or 50 cGy of Fe, Si, or Ti, or to gamma rays (from 25 cGy to 2 Gy) as a reference for the high LET (linear energy transfer) ions. Sample collections occur at either 12 hours, or 4 or 8 weeks after irradiation. Fecal samples are collected prior to irradiation and at the times listed after irradiation for microbial sequencing and metabolite analyses. Colon mucosa samples have been collected and will be processed for RNA isolation, epigenetic characterizations, or sections fixed to assess apoptosis and DNA damage in stem cells vs non-stem cell populations. Colon crypts isolated from samples collected at 4 and 8 weeks were also sorted to collect the stem and non-stem cells to perform population-specific gene expression and epigenetic state analyses. Progress to Date During the Spring 2017 run, we finished the last of our animal radiation exposures (0, 10, 25, or 50 cGy of 1 GeV/n Si and Ti ions). During the summer, we finalized sample collections and are currently in the process of performing the various analyses on the samples collected from all of the mice. For each run, a subset of the mice was sacrificed 12 hours after irradiation at BNL and the remaining mice were transported back to Texas A&M University. Animals sacrificed at 4 and 8 weeks were used to perform cell sorting in order to collect stem cells, daughter cells, and differentiated cells. Discussion: We anticipate that the beneficial diet containing fish oil and pectin will mitigate radiation-induced changes in the microbiota and their metabolites, essentially improving resilience of the luminal environment following radiation insult. With respect to the host, we anticipate that the epigenetic signature of colonocytes exposed to radiation will not be impacted to the same extent as those animals consuming diets that do not contain butyrate-yielding fibers or omega-3 fatty acids. The net outcome of these changes will be an increased responsiveness of colon adult stem cells to apoptosis induction and reduction in the transmission of undesirable epigenetic signatures to daughter cells.

	The initial observations from the cell sorting experiments support our original hypothesis. From that work, we have identified an ion-specific change in the proportion of adult colon stem cells. This response is present in all mice at 4 weeks after exposure but the change was less in mice terminated at 8 weeks after exposure. Importantly, the extent of the change at 4 weeks was not as great if the mice had been consuming the fish oil and pectin diet, and by 8 weeks, there was no difference between the sham irradiated mice and the irradiated mice if they were consuming the fish oil and pectin diet. We are in the process of performing the immunofluorescence analyses, the microbial sequencing analyses and the fecal short chain fatty acid analyses. As designed, these experiments will permit us to address the gaps in our understanding of how radiation impacts the colon, but importantly does so using a systems biology approach. This experimental paradigm enables us to address the gap concerning our understanding of host molecular, cellular, and tissue responses, along with changes that may occur in the microbiota in the colon. Therefore, these experiments will provide data relevant to resolving the risk of Radiation Carcinogenesis from Space Radiation.
Bibliography Type:	Description: (Last Updated: 05/28/2021)
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Articles in Peer-reviewed Journals	Little-Letsinger SE, Turner ND, Ford JR, Suva LJ, Bloomfield SA. "Omega-3 fatty acid modulation of serum and osteocyte tumor necrosis factor-alpha in adult mice exposed to ionizing radiation." J Appl Physiol (1985). 2021 Mar 1;130(3):627-39. https://doi.org/10.1152/japplphysiol.00848.2020 ; PMID: 33411639; PMCID: PMC7988793 , Mar-2021